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Experimental Model Derivation and Control of a Variable Pitch Propeller Equipped Quadrotor

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Abstract— **The aim of this article is to present a novel quaternion based control scheme for the attitude control problem of a quadrotor equipped with variable pitch propellers. A variable pitch propeller is a type of propeller system utilizes a mechanical mechanism to change the pitch of the rotor blades, while when applied to quadrotors it results in an over-actuated control system. The novelty of the article stems from: a) the proposal of an experimental model for variable pitch propellers and b) the novel proposed thrust and power consumption optimizations for the over-actuated quadrotor. Throughout the article, the merits of the proposed novel approach are being analyzed and discussed, while the efficiency of the variable pitch propellers are being evaluated by extended simulation and experimental results.**

I. Introduction

The area of Unmanned Aerial Vehicles (UAV) and especially the one of having the capability for Vertical Take-Off and Landing (VTOL) such as the quadrotor, has been in focus of research and development efforts, mainly due to their efficiency in accomplishing complex missions [1].

For achieving the desired performance, the trajectory generation problem is frequently being divided into two subproblems: a) the attitude problem and b) the translation problem, while as it has been proven [2, 3], these systems can be cascade interconnected. In these presented control approaches, the mechanical aspect of the control problem currently only utilizes fixed pitch propellers, where thrust and torque is generated by changing the rotational speed of the motors. For the examined case of a quadrotor-UAV, the position controller (translation) is generating reference attitude set–points for the attitude controller and thus the control problem of quadrotors has been confronted using several different approaches from research–leading teams worldwide, with famous works to include linear [4–6], and nonlinear controllers [7–11], however none of these have examined the case of redefining the fundamental mechanical constraints of a quadrotor by changing the fixed pitch propellers into variable pitch propellers.

Although, it has been proved that the aforementioned control strategies manage to stably navigate a quadrotor, the problem of designing optimized controllers that will be able to: a) provide fine–smoothed control actions for attitude stabilization and trajectory tracking, b) make use of model–knowledge for more accurate navigation, c) preserve robustness against sudden and unpredictable external disturbances, is still an open challenge. Moreover, one of the constrains that the control engineers are facing, when dealing with the attitude problem is still a fundamental problem in dynamics, mainly due to the fact that finite rotation of a rigid body does not obey the laws of vector addition, e.g. commutativity, while the attitude characteristics of the rigid body cannot be extracted by integrating the body's angular velocity.

However, when working with rotations, whether it's estimators or controllers, there has been one approach utilized more than any other: the Newton–Euler equations [12], which is able to describe the combined translational and rotational dynamics of the rigid body. Although, this modeling approach is considered a fundamental one, it still has three drawbacks: a) it is solely based on Euler angles, which have the merit of being intuitive, but per definition these angles cannot define certain orientations as it suffers from singularities that result in "gimbal lock" [13], which is the problem of loosing of one degree of freedom in a three-dimensional space that occurs when two of the rotational axes align and locking together, b) it is very computationally expensive, since calculating sines and cosines takes a lot of performance and can very fast become unmanageable especially if it is implemented on low cost hardware, and c) when designing estimators or controllers it is often necessary to utilize the Jacobian of the system states and during these calculations some times all the matrix elements will have one or more sines or cosines to compute, which quickly can overwhelm the system.

The other challenge control engineers face is the limitations of the quadrotor itself. In the case of utilizing fixed pitch propellers, the system can only react as fast as the engines can accelerate and decelerate their rotational speed. The efficiency of the overall system is only as good as the motor-propeller combination, with no room for changes except applying more voltage for a greater rotational speed. However, if the motor rotates too fast, an effect known as the blade stall comes into effect that dramatically lowers the performance of the propeller, since it reduces the lift produced. Due to the fact of bounded increase of the propeller's rotational speed, another approach can be considered by changing the pitch of the propeller, as in this case it is possible to have complete control over the thrust generated and the power consumed by adding another degree of freedom and transforming the quadrotor into an over-actuated system.

The novelty of this article stems from: a) theoretically and experimentally developing and evaluating novel models for variable pitch propellers, b) optimizing the variable pitch-

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thrust configurations, and c) evaluating the performance of the overall variable pitch quadrotor. To the authors best knowledge, this is the first time that an integrated full quaternion controller, based on theoretical and experimental variable pitch propeller models, have been proposed, as in the relative scientific literature: a) only a few references are available utilizing quaternions controllers for the attitude stabilization problem as it has been described in [14], [15], and the references there in, and b) scientific approaches for variable pitch quadrotors include the works in [16–18], but no on board controller and only partially complete models for thrust and power consumption have been developed.

The rest of the article is structured as it follows. In Section II the theoretical and experimental variable pitch propeller model derivations are being presented, while in Section III the optimization scheme for the pitch and thrust selections is being established. In Section IV the novel full quaternion based control scheme for variable pitch propellers is depicted, while in Section V extended simulation results for proving the efficiency of the proposed scheme are being presented. Finally, in Section VI the conclusions are drawn.

II. Variable Pitch Propeller Experimental Model Derivation

A variable pitch propeller is a type of propeller system utilized on out-runner motors, which have a mechanical mechanism to change the pitch of the rotor blades as opposed to their fixed pitch counterparts. Having this ability gives the system three new degrees of freedom: a) thrust can be directed in both upward and downward directions utilizing positive and negative pitch, b) the ability to control the power consumption by utilizing the optimal combination of pitch and rotational speed, and c) changing pitch and rotational speed to maximize the thrust response.

A. Theoretical Model

For consistency and simplicity of finding the variable pitch propeller's experimental model, a simplified theoretical model will be derived. A representation of a rotating propeller is presented in Figure 1, where the airflow is assumed to be laminar and the air in front of the propeller is stationary.

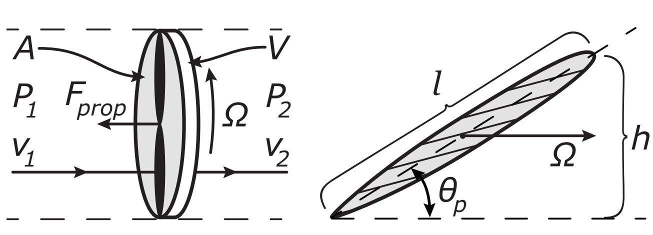

Fig. 1. Pressures and wind velocities generated by a propeller is presented in the left sub figure, where P_{1-2} are pressures, v_{1-2} are velocities, A is the area and V is the volume of one propeller revolution, Ω is the rotational speed and F_{prop} is the generated thrust, while in the right figure a cross section of a propeller is presented, where l is the width, h is the height relative the rotational direction and θ_p is the pitch angle of the blade.

From these assumptions and the Bernoulli's Principle, presented in (1):

$$\frac{v^2}{2} + gz + \frac{P}{\rho} = \text{const.}, \tag{1}$$

the change of airspeed and pressure from before to after the propeller is derived in (2, 3):

$$\frac{v_1^2}{2} + \frac{P_1}{\rho} = \frac{v_2^2}{2} + \frac{P_2}{\rho} \tag{2}$$

$$v_1 \approx 0 \Rightarrow \frac{v_2^2}{2} = \frac{P_1 - P_2}{\rho}, \tag{3}$$

while the force generated by the differential pressure can be denoted as (4):

$$\Delta P = \frac{F}{A} \Rightarrow F_{prop} = \frac{A\rho}{2} v_2^2 \tag{4}$$

To derive the airspeed after the propeller, it is assumed that one revolution of the blades moves an air-mass with cylindrical volume V in every revolution and it has a rotational speed Ω, resulting in the airflow Q and airspeed u_2 presented in equations (5, 6) as:

$$Q = V \cdot 2\pi\Omega = 2Ah \cdot 2\pi\Omega = 4\pi Al \sin(\theta_p)\Omega \tag{5}$$

$$v_2 = \frac{Q}{A} = 4\pi l \sin(\theta_p)\Omega \tag{6}$$

The resulting force generated F_{prop} is presented as it follows:

$$F_{prop} = 8\pi^2 A l^2 \rho \sin^2(\theta_p)\Omega^2 = A_F^{\text{thr}} \sin^2(\theta_p)\Omega^2 \tag{7}$$

with $A_F^{\text{thr}} = 1.8 \cdot 10^{-3}$, however it should be noted that this value will not be used in the final result. This model was designed to provide the general characteristics of the propellers for experimental validation.

This model derivation is simpler than the model derived by [18], however this model will prove more accurate as will be discussed in the experimental validation.

B. Experimental Setup

The experimental setup for constructing the model of the variable pitch propeller system, is being depicted In Figure 2. As it can be observed, there is a rod going though the center of the electric motor, connecting the variable pitch mechanism to a servo-actuator controlling the pitch. Moreover, there is an angle sensor, which measures the angle of the servo-actuator, and a sensor that measures the rotational speed of the motor attached to the side of the assembly.

The electric motor utilized is the AXI 2212/20 EVP and is of a brushless design, which implies that it is a 3-phase Alternating Current (AC) Synchronous Motor and driving it directly from a Direct Current (DC) voltage, as with a normal brushed motor, will not work because it needs three voltages with 120° phase shift from each other. Converting DC to the three signals is handled by the Electronic Speed Controller (ESC) that effectively works as an 3-phase inverter that controls the frequency of the signals in order to control the rotational speed, since this frequency is directly proportional to the rotational speed of the motor.

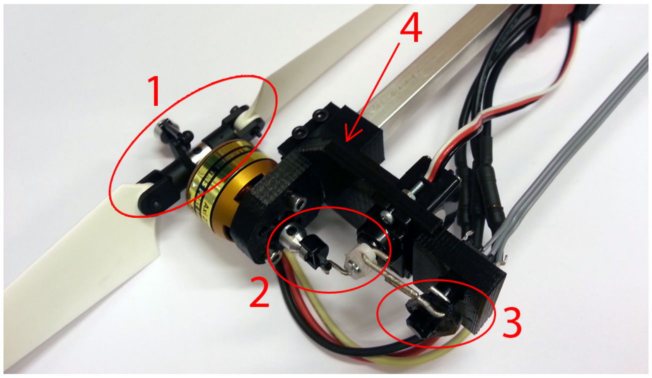

Fig. 2. A variable pitch propeller mounted to the experimental base. (1) is the Variable Pitch assembly, (2) is the servo actuator's rotating push rod, (3) is the angle measurement assembly and (4) is the printed plastic holding frame.

The complete experimental set–up to measure thrust, rotational speed, pitch angle and power consumption is presented in Figure 3. This system has been designed to ensure simplicity of parts and simple calculations. It consists of the motor and sensor assembly on one of the square tubes, with the other tube pressing against a scale to measure thrust and a weight to counteract the weight of the motor and the sensor assembly. The distance from the rotational point to the motor assembly and to the scale is the same, hence the mass measured directly corresponds to the thrust being generated. In order to guarantee that the tilting system would not introduce errors, it was suspended in low friction ball-bearings, while it has been assumed that negligible counter torque is produced in the rotation point.

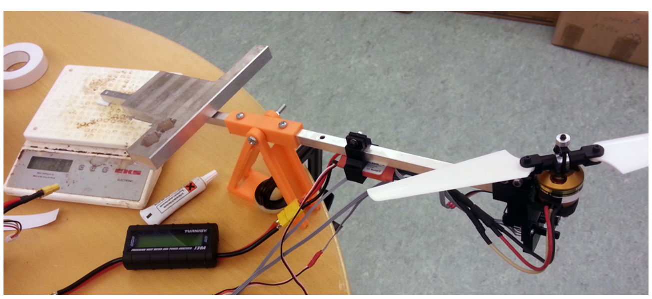

Fig. 3. The variable pitch propeller and sensor assembly mounted to the complete experimental setup.

C. Experimental Results and Model Derivation

For the following model derivations, the corresponding data sets have been gathered by utilizing grid based measurements with a pitch step of 0.077rad and 10% increments in throttle.

Static Models: For deriving the theoretical constant A_F^{thr} from (7), denoted as A_F^{exp}, the experimental data–points depicted in Figure 4 have been utilized. More specifically, a Least-Squares plane fit has been applied, which produced the corresponding mesh–grid, depicted at the same Figure 4, with $A_F^{\text{exp}} = 1.788 \cdot 10^{-5}$. As it can be observed, the data-points and corresponding model matches very well with very small errors with an RMSE of 0.17 N. This also shows, in contradiction with [18], that the propellers can produce great force at higher pitch angles. This, most likely, comes from the fact that the Newtonian component of the generated thrust overtakes the lift generated laminar flow of air when the blades go into blade stall. However, for quick maneuvers is it possible to use these high pitch angles.

The derivation of the the power consumption model has been executed by examining the power consumption data–points versus the pitch angle and the rotational speed as it can be depicted in Figure 5. Under a constant rotational speed, a general trend was found that can be mathematically formulated as it follows:

$$P(\theta_p, \Omega) = \left(A_P + B_P \sin^5 \theta_p\right) \Omega^2 \tag{8}$$

with A_P and B_P being general constants. This $P(\theta_p, \Omega)$ relation was proved close to a perfect match with very small error deviations with an RMSE of 3.8 W as it can be observed in Figure 5, by overlaying the generated plane over the measurements, with $A_P = 3.233 \cdot 10^{-5}$ and $B_P = 2.127 \cdot 10^{-3}$.

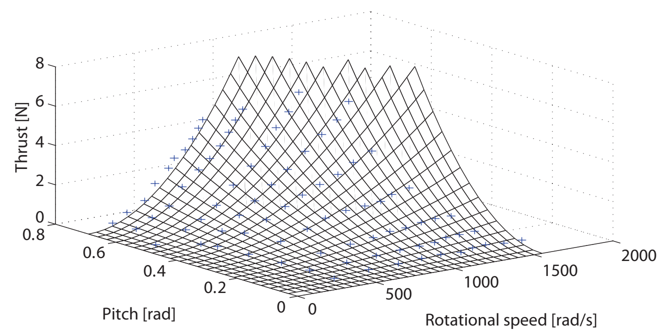

Fig. 4. Model fit of 90 static thrust measurements. The crosses denote measurements and the mesh is the plane fit.

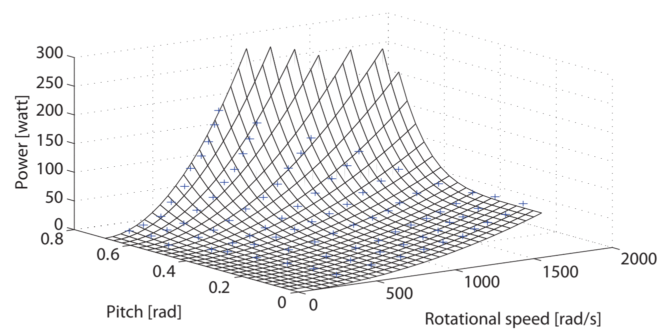

Fig. 5. Model fit of 90 static power consumption measurements. The crosses denote measurements and the mesh is the plane fit.

Transient Models: To derive the transient models, step responses of pitch and rotational speed, with respect to constant step reference changes, were examined. In both cases a first order response fit was successfully derived with

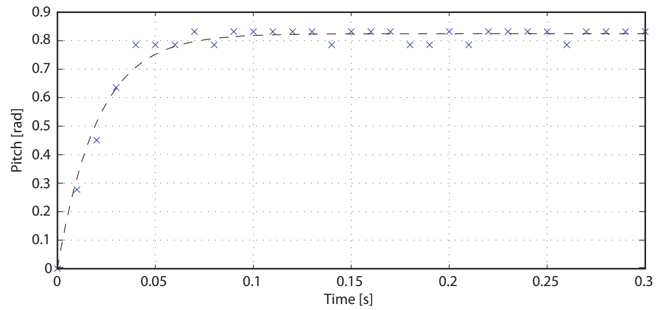

Fig. 6. Transient model fit of pitch step response. Crosses denotes measurements and dashed line is model fit.

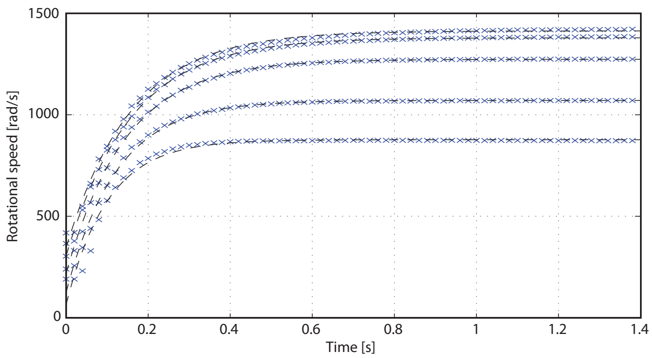

Fig. 7. Transient model fits of rotational speed step responses. Crosses denotes measurements and dashed line is model fit.

very small error deviations, as depicted in Figures 6 and 7, with the resulting transfer function presented in equations (9) and (10), where the control signal has been bounded by $-1 \leq u_\theta \leq 1$ and with a time constant of $\tau_\theta = 0.0206$:

$$\theta_p(s) = \frac{\theta_s}{1 + \tau_\theta s} \tag{9}$$

$$\theta_s = \theta_{p,max}\ u_\theta$$

$$\Omega(s) = \frac{\Omega_s(\theta_p, u)}{1 + \tau_\Omega(\theta_p) s} \tag{10}$$

with $\tau_\Omega(\theta_p)$ and $\Omega_s(\theta_p, u)$ the time constant and the peak value of the rotational speed respectively. However, at different pitches, the $\tau_\Omega(\theta_p)$, $\Omega_s(\theta_p, u)$ constants change, as it can be observed from the experimental data presented in Figure 7, and thus two second order models for the time constant and the peak value of the rotational speed have been further elaborated as it follows:

$$\tau_\Omega(\theta_p) = A_\Omega + B_\Omega \theta_p^2 \tag{11}$$

$$\Omega_s(\theta_p, u) = \left(A_u + B_u \theta_p^2\right) u_\Omega \tag{12}$$

with the following parameters' values:

$$A_\Omega = 0.1649,\ B_\Omega = -0.1029$$
$$A_u = 1.636 \cdot 10^3,\ B_u = -1.974 \cdot 10^3$$

and with the corresponding control signal bounded in $0 \leq u_\Omega \leq 1$. These models, have been experimentally proven to be sufficient to capture the general response of the variable pitch propellers as it can be depicted in Figure 8. The two control signals, u_θ and u_Ω, were chosen to always be relative to the maximum.

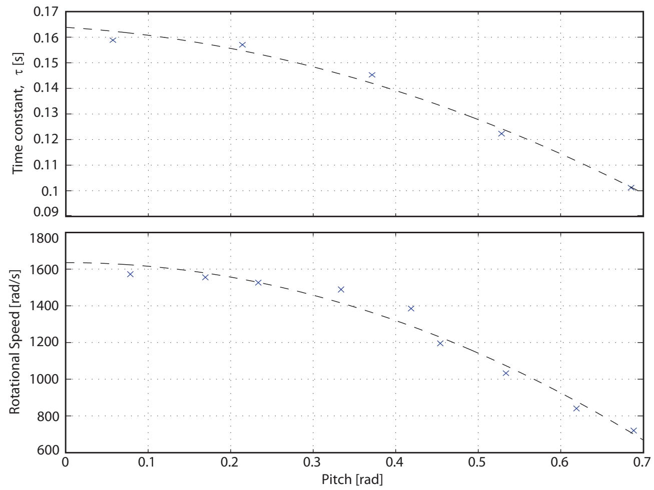

Fig. 8. Top figure depicts the time constant of rotational speed as a function of pitch angle and the bottom figure depict the maximum rotational speed as a function of pitch angle respectively.

To conclude this subsection, novel theoretical models have been combined with experimental data to produce thrust and power consumption models for variable pitch propellers as a function of the rotational speed and the propeller pitch. In the sequel these models will be utilized in Section III in order to optimize the operation of quadrotor: a) from a power consumption and b) a force derivative (fast response) point of view. Finally, in Section IV a novel full quaternion based attitude control scheme, for a variable pitch quadrotor, will be presented.

III. Optimizations

Since variable pitch propellers are over actuated systems, where many combinations of pitch and rotational speed can produce the same thrust, it is desirable to find optimal operating areas depending on different requirements and conditions. In such a consideration, there are mainly two optimizations that can be made: a) minimize power consumption or b) change the thrust vector as fast as possible. In this section these two conditions will be examined, resulting in general guidelines for throttle and pitch configurations depending on which optimization constraint is required.

A. Power Consumption

The need of optimization of power consumption is evident when a low power flight is required. This situation can be observed during hovering, slow movements or when the time for attitude convergence is not critical. Hence, it is desired to minimize power consumption depending on the rotational speed and the pitch of the propeller as the first part of (13) describes. The minimization of the power consumption, is equivalent to the maximization of the thrust to power ratio.

By utilizing equations (7, 8) the thrust to power ratio is described in the second part of equation (13) as:

$$\min_{\theta_p,\ \Omega} P(F,\theta_p,\Omega) = \max_{\theta_p} \frac{F}{P} = \\ = \max_{\theta_p} \frac{A_F^{\exp}\sin^2\theta_p}{A_P + B_P\sin^5\theta_p} \tag{13}$$

At this point, it should be noted that in the resulting equation (13) there has been no dependence on the rotational speed. This observation shows that the optimal thrust to power ratio is independent of the rotational speed and one or more pitch angles will be able to produce optimal results. The resulting thrust to power ratio dependence is presented in Figure 9, where at $\theta_p = 0.414$ rad is the optimal pitch for the best thrust to power ratio. This figure also indicates that the best achievable thrust to power ratio is 52.9 mN/W or approximately 5.39 grams/W. This is also in contradiction with [18] since they showed that there should be a dependence on the rotational speed as well, however the experiments showed this to be false or have such a small impact that it is lost in the other major components of the power consumption.

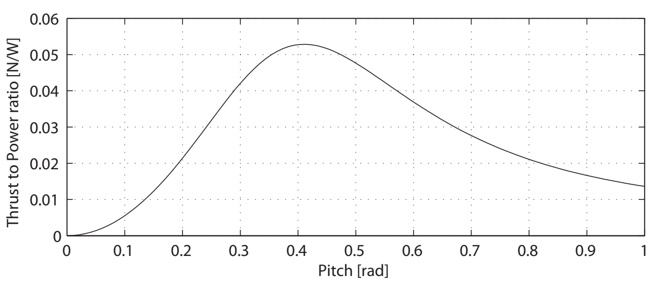

Fig. 9. Thrust to Power ratio as a function of pitch.

B. Force Derivative

In the case that quick and agile moves are required, the quicker the response of the thrust is, the faster the torque and the acceleration can be changed. For achieving an optimal configuration, it is necessary to maximize the derivative of the equation (7) as depicted in equation (14), while the peak value should come with the smallest time delay with respect to the reference change, in order to guarantee a fast thrust response:

$$\max_{\theta_p,\ \Omega} \dot{F}(\theta_p(t),\ \Omega(t)) \tag{14}$$

The force equation from equation (7) can be rewritten as two functions independent of each other, as it is presented in equation (15):

$$F(\theta_p(t),\Omega(t)) = A_F^{\exp}\sin^2(\theta_p)\Omega^2 = \\ = A_F^{\exp}\cdot F_\theta(\theta_p)\cdot F_\Omega(\Omega) \tag{15}$$

The derivation with respect to time gives a result with better properties for analyzing, as presented in equation (16).

$$\dot{F}(\theta_p(t),\Omega(t)) = A_F^{\exp}\Big(F_\theta(\theta_p)\cdot\dot{F}_\Omega(\Omega) + \\ + \dot{F}_\theta(\theta_p)\cdot F_\Omega(\Omega)\Big) \tag{16}$$

With respect to the equation (16) two cases can be considered: a) where the pitch is assumed to be constant and b) where the rotational speed is assumed constant. This will effectively reduce the complexity from a maximization problem in two dimensions to a problem in one dimension because of the corresponding derivative being zero, while the utilization of this approach will also allow to calculate how much effect each of the cases have on the derivative. However, special attention should be taken if the derivatives' peaks are close to each other, as in this case the assumption of two one dimensional problems will not hold and a combined equation must be examined.

a) Constant pitch: The assumption of a constant pitch cancels the pitch derivative because $\dot{F}_\theta(\theta_p) = 0$ and only the rotational speed derivative, $\dot{F}_\Omega(\Omega(t)) = 2\Omega\dot{\Omega}$, has to be examined, with the maximum value of:

$$\max\left\{\dot{F}(\theta_s,\Omega(t))\right\} = A_F^{\exp}\sin^2(\theta_s)\frac{\Omega_s^2}{2\tau_\Omega}$$

is found at $t_\Omega = \ln(2)\tau_\Omega$.

b) Constant rotational speed: Using the previous technique and constant rotational speed, instead of a constant pitch, only the pitch derivative, $\dot{F}_\theta(\theta_p(t)) = \sin(2\theta_p)\dot{\theta}_p$, has to be examined and the maximum value of:

$$\max\left\{\dot{F}(\theta_p(t),\Omega_s)\right\} = A_F^{\exp}\sin(\theta_s)\theta_s\frac{\Omega_s^2}{2\tau_\theta}$$

is found at $t_\theta = \frac{2}{3}\tau_\theta$.

Conclusions: To compare the results from the previous equations, simplifying approximations are made as:

$$\sin(\theta_s)\theta_s \approx \sin^2(\theta_s),\ \ln(2) \approx \frac{2}{3}$$

and by using these approximations and removing the common parts, two very clear results emerges. a) as presented in equation (17), the greatest peak in derivative is provided by the transfer function, from equations (10, 9) with the smallest time constant and b) the time during the peak value is being reached, as presented in equation (18), is also provided by the transfer function with the smallest time constant.

$$\frac{1}{\tau_\theta} >> \frac{1}{\tau_\Omega} \tag{17}$$

$$t_\theta << t_\Omega \tag{18}$$

In both cases changing pitch and keeping rotational speed constant provides the fastest thrust response because of the servo actuators' time constant being approximately one order of magnitude smaller than the time constant of rotational speed. This shows that for the fastest thrust response, changing pitch provides the best result and due to the fact that the peaks resides far apart in time, the assumption with two distinct cases holds.

IV. Variable Pitch Propeller Quaternion based Control

Due to size constraints the original fixed pitch controller will not be presented. The original controller with an in depth analysis of quaternions and the initial formulations can be located in [18, 19].

To compare the differences between fixed pitch propellers and variable pitch propellers, the simulation model and controller is the same as used in [19], but instead of using the assumption that control signal to torque is an identity matrix the real models, deduced in Section II, are utilized, resulting in a different control action, where u has the same limits as u_θ and u_Ω in equations (9) and (12) depending on if the pitch or rotational speed is controlled. Based on this formulation, the generated torque can be expressed as in the following two cases of differential pitch and differential trust.

$$\tau_p(\theta_d) = l_a A_F \left(\sin^2(\theta_t + \theta_d) - \sin^2(\theta_t - \theta_d)\right) \Omega_t^2$$

In the differential pitch formulation θ_t is the bias pitch for thrust, θ_d is the controlled differential pitch as $\theta_d = \theta_{max} u$ and l_a is the length from the motor to the rotational center.

$$\tau_\Omega(\Omega_d) = l_a A_F \sin^2(\theta_t) \left((\Omega_t + \Omega_d)^2 - (\Omega_t - \Omega_d)^2\right)$$

In the differential thrust formulation Ω_t is the bias throttle and Ω_d is the differential throttle as $\Omega_d = \Omega_{max} u$.

Based on these formulations and analysis, the resulting novel proposed quaternion based attitude control scheme can be presented in a block diagram formulation as in Figure 10. Depending on the optimization cost and the needs of the mission, proper optimal running for the pitch angle or the thrust can be directly achieved to obtained a significantly improved performance, as it will be also indicated by the simulation results presented in Section V.

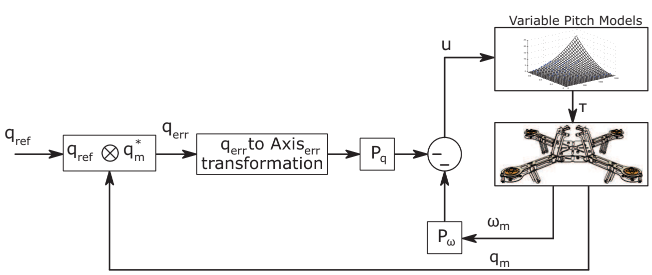

Fig. 10. Block diagram of the full non-linear P^2 quaternion based control scheme with variable pitch propellers.

V. Simulations

All simulations have been carried out on the nonlinear quadrotor model presented in [10], which takes into account a wide set of the aerodynamic forces and moments acting on the system, including the hub and friction forces, the rolling moments and –up to some extent– the variations in the aerodynamic coefficients, due to the motion of the quadrotor inside the atmosphere. The parameters of the quadrotor model have been set as $I_{xx} = I_{yy} = 6.5 \cdot 10^{-4} kg \cdot m^2$, $I_{zz} = 1.2 \cdot 10^{-3} kg \cdot m^2$ and $l_a = 0.25m$ where the aforementioned values correspond to the CAD model analysis presented in [19]. The gains of the nonlinear P^2 controller have been set as: $P_q = 4$ and $P_\omega = 1.2$ for the case of fixed pitch and to $P_q = 3$ and $P_\omega = 0.6$ for the case of variable pitch, the change from [19] comes from the control signal now being bounded to relative signals and not absolute torque.

Utilizing this system, all the non-linear effects from control signal to torque are included and the real response of the system can be simulated, analyzed and characterized. In the sequel, the response to a step reference was examined in two cases: a) by utilizing fixed optimal pitch and varying the rotational speed and b) by utilizing fixed rotational speed at about 80% throttle and varying the pitch of the propellers. The effect of fixed pitch versus variable pitch on rise time is evident in Figure 11, where the variable pitch case is approximately twice as fast as the fixed pitch, without any increase in overshoot.

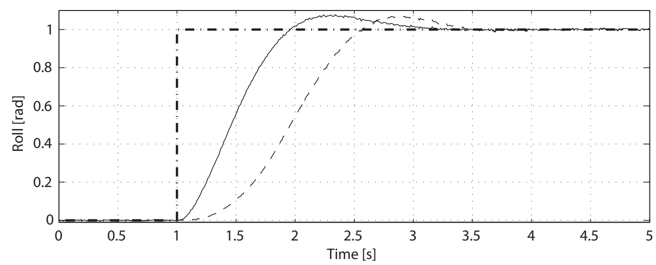

Fig. 11. The roll angle response during a step reference change at t = 1 second. Dashed line is the fixed pitch case, solid line is variable pitch and dash-doted line is reference. Graph have been indicated in radians for intuitive display.

It was predicted in Section III that the force derivative for the variable pitch case should be approximately one order of magnitude larger than the fixed pitch case and from Figure 12 it is evident that the force derivative is much larger for the variable pitch case over the fixed pitch case, as predicted. This makes the fast change of roll in Figure 11 possible, as a very large amount of torque is initially produced, as it can be observed in Figure 12, for enabling the quadrotor to start rotating.

Moreover, from Figure 12 it is quite evident that the thrust being generated is way outside the maximum from what was measured in the experiments and this comes from the fact that the propellers have built up inertia and rotational speed while being at low pitch. When the pitch change is executed the system responds with enormous thrust until the rotational speed decreases, as presented in Figures 13 and 14. This process takes approximately 200 ms, but during these 200 ms the system can produce thrust much greater than the static measured thrust.

VI. Conclusions

In this article a novel full quaternion based attitude controller for a quadrotor equipped with variable pitch propellers, using models from theoretical derivations and confirmed with experimental data has been presented. Extended simulations and experimental results have been presented that

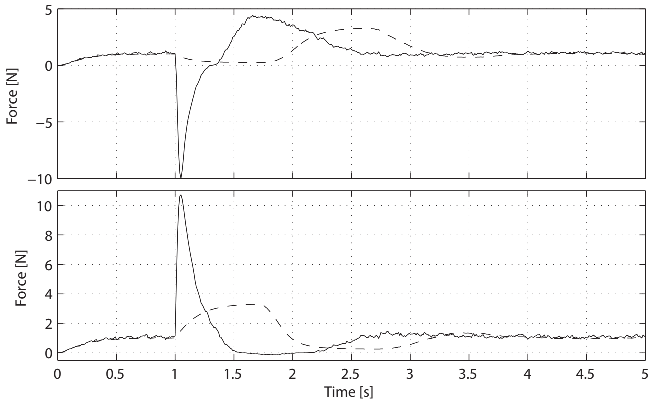

Fig. 12. The force generated by two opposite motors during a step reference change at t = 1 second. Dashed line is the fixed pitch and solid line is variable pitch.

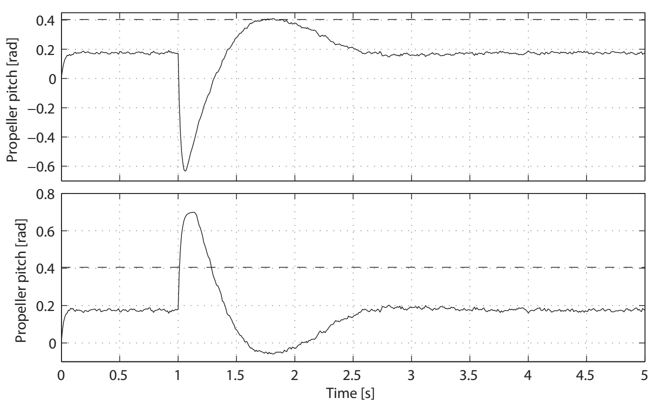

Fig. 13. The propeller pitch of two opposite motors during a step reference change at t = 1 second. Dashed line is the fixed pitch and solid line is variable pitch.

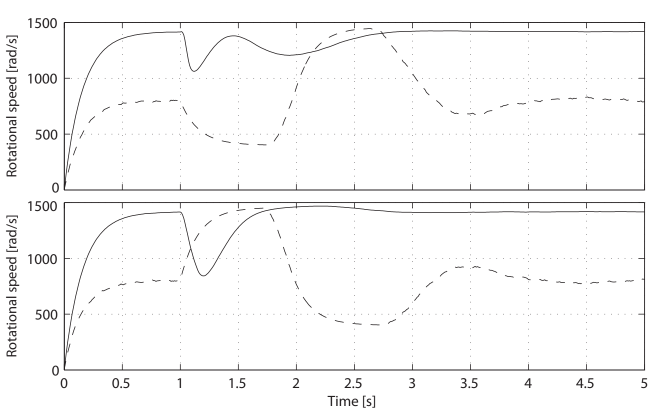

Fig. 14. The rotational speed of two opposite motors during a step reference change at t = 1 second. Dashed line is the fixed case and solid line is variable pitch.

proves the efficiency of the proposed propeller assembly and suggested control scheme.

Future work will include experiments on a full variable pitch quadrotor, confirming the simulated responses and designing a controller which can take into consideration the effects of inverted flight, while adjusting pitch and navigation to compensate accordingly.

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